

Electronics on Wheels

Today, car manufacturers use electronics not just for safety and security, networking and energy management but also for entertainment, driver assistance and passengers' comfort. In short, they are using electronics to boost the 'user experience' holistically. Here are some brilliant examples



Car dashboard with GPS panel

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1. Visteon e-Bee

Intelligent mobility. Visteon's e-Bee concept car gathers and disseminates useful driving-related data, acts as part of a wider mobility-enabling network, and integrates technology and design that is connected, simple and flexible.

Camera-enhanced cockpit concept. It assists the driver by providing automatic personalisation, pedestrian sensing, traffic sign recognition and even intruder alerts.

Human Artificial Bayesian Intelligence Technology (HABIT). It uses artificial intelligence to learn the driver's habits and creates a truly personalised

experience based on this information. It also features cloud-centred profile storage.

Pressure-activated smart panel. This smart panel detects the presence of the user's finger but does not activate until a small amount of pressure is applied. This 'touch with intent' capability is enhanced with visual and audible feedback, and will be expanded on future implementations.

Head-up display solution. Developed in partnership with BAE Systems, it delivers industry-leading display image quality in a compact and efficient optical design.

Global digital radio solution. It provides software-reconfigurable mobile reception across multiple digital radio standards.

2. Cadillac User Experience

Contextual design. The Cadillac User Experience (CUE) has been designed ground up, based on an understanding of consumer usage and wishes. The designers classified users into eight distinct personas to understand their needs and develop features. Then, they built an extensive design and engineering specification manual that included more than 1500 pages of storyboards, functions and program wire-frames. They also created HMI state charts that described the interactive behaviour of the system to be programmed.

Response touch-screen. A 20.3cm (8-inch) capacitive touch screen allows swipe and pinching gestures to move and resize items on the main screen.

Haptic feedback. The panel gives users a pulsating sensation when they select menu items.

Proximity sensing. This technology detects the presence of users and minimises the display when not in use, thereby conserving power.

Built-in 3D navigation. It includes turn-by-turn natural speech recognition and auto-fill location input. It also provides map-integrated weather reports.

Open platform. The Linux-based CUE platform is open to developers, so they can develop new applications for Cadillac.

3. Mercedes-Benz

Cross-traffic assist. Mercedes-Benz' BAS Plus technology reduces rear-end